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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBBT102 -

FUNDAMENTALS OF ENZYMOLOGY AND RADIOBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-Answer all questions-Each question carries 1 marks

1. What is polarity? [K] / [CO1]
    a. Definition- 1 mark
2. What is the dilution factor? [K] / [CO2]

    Explanation- .5 mark, Equation-.5 mark
3. Define absolute specificity of an enzyme with an example. [K] / [CO3]

    Definition- .5 mark, Example- .5 mark
4. Write down Michael - Menten equation. [K] / [CO4]

    Equation- .5 mark, Explanation of terms- .5 mark
5. What is a coenzyme? [K] / [CO3]

    Definition – .5 mark Explanation – .5 mark
6. Define oxidoreductases with an example. [K] / [CO3]

    Definition – .5 mark Example – .5 mark
7. Define isoenzymes [K] / [CO4]

    Definition – .5 mark Example – .5 mark
8. How does temperature affect enzyme activity? [K] / [CO4]

    Explanation (2 points) – 1 mark
9. How does Concentration of enzyme affect enzyme activity? [K] / [CO4]

    Explanation – (2 points) 1 mark
10. Give any two properties of beta rays [K] / [CO6]

    2 properties – 1 mark (.5 mark each)

Part B-Answer any 4 questions-Each question carries 2 marks

11. What is the formula for calculating molality? [K] / [CO2]

    Formula – 1 mark Explanation of terms – 1 mark
12. Give two properties of buffers. [K] / [CO2]

2 properties –2 marks (1 mark each)

13. What is the principle of scintillation counter? [U] / [CO6]

    Principle – 2marks

14. How do activators affect enzyme activity? [U] / [CO4]

    Explanation – (4 points) 2 marks

15. Describe lyases with an example. [K] / [CO3]

    Description – 1 mark example – 1 mark

16. Explain uncompetitive inhibition [U] / [CO5]

    Explanation – 2 marks

Part C-Answer any 3 questions-Each question carries 4 marks

17. Write notes on pH and pH scale [U] / [CO2]

    pH- definition – 1 mark ,Explanation – 1 mark, pH scale – 2 marks

18. Explain the structure and working of pH meter. [A] / [CO2]

    Structure – 1 mark Figure – 1 mark working – 2 marks

19. Describe enzyme inhibition [A] / [CO5]

    Definition - 1 mark, Competitive – 1 mark, Uncompetitive – 1 mark, Non-competitive – 1 mark

20. Describe Geiger Muller counter [A] / [CO6]

    Explanation – 1 mark Principle – 1 mark Schematic diagram – 1 mark Working – 1 mark

21. Elaborate on autoradiography [A] / [CO6]

    Definition – 1 mark Principle – 1 mark working – 2 marks

Part D- Answer any 2 questions- Each question carries 10 marks

22. Elaborate on acids and bases [U] / [CO2]

    Introduction – 2 marks4 theories –4marks (1 mark each), example for each – 4 marks (1 mark each)

23. Describe enzyme regulation. [A] / [CO3]

    Enzyme regulation- definition-1 mark, Major types –4 marks (1 mark each) Explanation – 5 marks

24. Elaborate on the applications of various radioisotopes in biology [E] / [CO6]

    Introduction – 2 marks, carbon, phosphorus, iodine, sulphur –8 marks (2 marks each)

25. Describe the mechanism of enzyme action [A] / [CO3, CO5]

    General mechanism – 2 marks, Lock and key – 3 marks, Figure – 1 mark, Induced fit – 3 marks ,Figure – 1 mark